

Leon Suarez-Rodriguez

Universidad Industrial de Santander • Bucaramanga, Colombia

• [Scholar](#) • [LinkedIn](#) • leonsuarez24@gmail.com

Education

Universidad Industrial de Santander Civil Engineering. Cum Laude	Bucaramanga, Colombia April 2023
Universidad Industrial de Santander Master's Degree in Computer and Systems Engineering	Bucaramanga, Colombia September 2025
Universidad Industrial de Santander PhD's Degree in Computer Science	Bucaramanga, Colombia Studying

Experience

Universidad Industrial de Santander Assistant Researcher in Project 3735 • Design and implementation of a preliminary deep learning model for action recognition. • Preliminary training of deep learning architecture for action recognition.	Bucaramanga, Colombia June 2023 – July 2023
Universidad Industrial de Santander Assistant Researcher in Project 3925 • Collect state-of-the-art information and techniques related to different problems of poststack seismic images. • Mathematically model the degradation of seismic images using image and signal processing techniques. • Implement different image processing techniques that allow degrading seismic images post-stacking. • In a controlled manner, including noise, blur, transformations and others. • Participate in the organization of activities for social appropriation of knowledge and dissemination of the project.	Bucaramanga, Colombia August 2023 – December 2023
Universidad Industrial de Santander Assistant Researcher in Project 8091 • Review the state of the art to determine interpolation strategies using neural networks based on continuous representation and coordinates. • Formulate mathematically the problem of spatial data interpolation with neural networks. • Implement different neural network models and conduct experiments to validate the interpolation capability of each method. • Validate the interpolation results with gravimetry field data from the Cerro Machín Volcano. • Prepare the gravimetry interpolation chapter of reports o2r4 and o2r5. • Present the results of project 8091 at national or international scientific events. • Participate in the implementation of activities for social appropriation of knowledge and/or training of human talent.	Bucaramanga, Colombia September 2025 – December 2025
Universidad Industrial de Santander Assistant Researcher in Project 8091 • Integrate interpolation algorithms based on deep learning into a repository and keep it updated. • Conduct technical meetings for the development of software that integrates the developed algorithms, with the purpose of defining the computational requirements and development languages for the creation of the graphical interface. • Consolidate the final versions of all technical reports for Objective 2 of Project 8091: O2R1, O2R2, and O2R3.	Bucaramanga, Colombia January 2026 – July 2026

- Prepare one (1) final technical report with the results of the validation of the deep learning model for the interpolation of punctual geophysical measurements, considering different sampling rates and topographic conditions.
- Consolidate a manuscript in a publishable version as author or co-author incorporating the results of Objective No. 2 - 8091.
- Organize and execute social appropriation of knowledge activities related to scientific events, workshops, seminars, and knowledge transfer strategies to the academy and/or human talent training.
- Participate in dissemination events and activities related to the social appropriation of knowledge.
- Perform other activities determined by the contract supervisor related to the fulfillment of contractual obligations and the purpose of the contract.

Publications

- P. Goyes-Peñafiel, L. Suárez-Rodríguez, C. V. Correa and H. Arguello, "GAN Supervised Seismic Data Reconstruction: An Enhanced Learning for Improved Generalization," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1-10, 2024, Art no. 5921910, doi: 10.1109/TGRS.2024.3434474.
- L. Suarez-Rodriguez, R. Jacome and H. Arguello, "Highly Constrained Coded Aperture Imaging Systems Design Via a Knowledge Distillation Approach," 2024 IEEE International Conference on Image Processing (ICIP), Abu Dhabi, United Arab Emirates, 2024, pp. 3993-3999, doi: 10.1109/ICIP51287.2024.10648100.
- R. Gualdrón-Hurtado, R. Jacome, L. Suarez, E. Martinez and H. Arguello, "Improving Compressive Imaging Recovery via Measurement Augmentation," ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Hyderabad, India, 2025, pp. 1-5, doi: 10.1109/ICASSP49660.2025.10888734.
- E. Martinez, L. Suarez, R. Gualdrón-Hurtado, R. Jacome and H. Arguello, "Compressive Imaging Reconstruction via Conditional Diffusion Model With Augmented Measurements," ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Hyderabad, India, 2025, pp. 1-5, doi: 10.1109/ICASSP49660.2025.10889114.
- R. Jacome, L. Suarez, R. Gualdrón-Hurtado, L. Gonzalez and H. Arguello, "Learning to Reconstruct Signals With Inexact Sensing Operator via Knowledge Distillation," ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Hyderabad, India, 2025, pp. 1-5, doi: 10.1109/ICASSP49660.2025.10887652.
- L. Suarez-Rodriguez, R. Jacome and H. Arguello, "Distilling Knowledge for Designing Computational Imaging Systems," in *IEEE Transactions on Computational Imaging*, vol. 11, pp. 1306-1319, 2025, doi: 10.1109/TCI.2025.3612849.
- R. Jacome, R. Gualdrón-Hurtado, L. Suárez-Rodríguez, and H. Arguello, 'NPN: Non-Linear Projections of the Null-Space for Imaging Inverse Problems', in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- L. Suarez-Rodriguez, R. Jacome, R. Gualdrón-Hurtado, A. Mantilla-Dulcey and H. Arguello, "DICE: Diffusion Consensus Equilibrium for Sparse-view CT Reconstruction," 2025 IEEE 10th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP), Punta Cana, Dominican Republic, 2025, pp. 261-265, doi: 10.1109/CAMSAP66162.2025.11423960.
- R. Jacome, R. Gualdrón-Hurtado, L. Suarez-Rodriguez and H. Arguello, "UTOPY: Unrolling Algorithm Learning via Fidelity Homotopy for Inverse Problems," 2025 IEEE 10th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP), Punta Cana, Dominican Republic, 2025, pp. 176-180, doi: 10.1109/CAMSAP66162.2025.11423864.
- J. Barrios, S. Rodríguez, K. Fonseca, L. Suarez-Rodriguez and H. Arguello, "Knowledge Distillation for Single Pixel Imaging Classification," 2025 XXV Symposium of Image, Signal Processing, and Artificial Vision (ST-SIVA), Armenia, Colombia, 2025, pp. 1-5, doi: 10.1109/STSIVA66383.2025.11156497.

- K. Fonseca, L. Suarez-Rodriguez, A. Jerez, F. Gutierrez-Barragan and H. Arguello, "Single Snapshot Distillation for Phase Coded Mask Design in Phase Retrieval,"2025 IEEE International Conference on Image Processing (ICIP), Anchorage, AK, USA, 2025, pp. 2019-2024, doi: 10.1109/ICIP55913.2025.11084573.
- R. Gualdrón-Hurtado, R. Jacome, L. Suarez, L. Galvis and H. Arguello, "Deep Distillation Gradient Pre-conditioning for Inverse Problems,"2025 IEEE 10th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP), Punta Cana, Dominican Republic, 2025, pp. 166-170, doi: 10.1109/CAMSAP66162.2025.11423951.

Languages

- Spanish (Native)
- English (Professional working proficiency)